

## ELCT 361: Electromagnetics

### Solution for HW #3

(a) Gauss's Law: the total outward flux of the E-field over any closed surface in free space is equal to the total charge enclosed in the surface divided by  $\epsilon_0$ .

$$\int_V \nabla \cdot \vec{E}(\rho) dV = \frac{1}{\epsilon_0} \int_V \rho dV \rightarrow \oint_S \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0} \text{ or } \frac{q}{\epsilon_0}$$

$$(b) \oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \rightarrow \oint_S \vec{E} \cdot d\vec{S} = E_R(\hat{a}_R) \cdot (\hat{a}_R) 4\pi R^2 = \frac{Q(q)}{\epsilon_0}$$

$$E_R = \frac{Q}{4\pi\epsilon_0 R^2}, \text{ therefore, } \boxed{\vec{E} = \hat{a}_R \frac{Q(q)}{4\pi\epsilon_0 R^2}}$$

$$(c) \vec{D} = \epsilon_0 \vec{E} = \epsilon_0 \hat{a}_R \frac{Q(q)}{4\pi R^2} = \boxed{\hat{a}_R \frac{Q(q)}{4\pi R^2}}$$

$$(d) \vec{F}_q(q) = \hat{a}_R q Q \vec{E} = \boxed{\hat{a}_R \frac{q Q}{4\pi\epsilon_0 R^2}}$$

$$(e) V_{R\infty} = - \int_{\infty}^R \vec{E} \cdot d\vec{l} = - \int_{\infty}^R \hat{a}_R \left( \frac{Q}{4\pi\epsilon_0 R^2} \right) \cdot \hat{a}_R dR$$
$$= - \frac{Q}{4\pi\epsilon_0} \left( -\frac{1}{R} \right) \Big|_{\infty}^R = \frac{Q}{4\pi\epsilon_0} \left[ \frac{1}{R} - \frac{1}{\infty} \right]$$
$$= \boxed{\frac{Q}{4\pi\epsilon_0 R}}$$

2.

(a) considering for chapter 3-2-17 & 3-2-18,

According to Gauss's Law,

$$\oint_s \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0} = \frac{\rho_l L}{\epsilon_0} \rightarrow 2\pi r L \bar{E}_r = \frac{\rho_l L}{\epsilon_0}$$

$$\boxed{\vec{E} = \hat{a}_r \frac{\rho_l}{2\pi\epsilon_0 r}}$$

$$(1) \vec{D} = \epsilon_0 \vec{E} \rightarrow \vec{D} = \hat{a}_r \frac{\rho_l}{2\pi r} = \boxed{\hat{a}_r \frac{\rho_l}{2\pi r}}$$

$$(2) \vec{E}_A = \hat{a}_r \frac{\rho_l}{2\pi r \epsilon_0} \quad (\phi, z, \text{ is not considered, respectively})$$

$$A(2m, \frac{\pi}{2}, 0), \quad r=2m.$$

$$= \hat{a}_r \frac{10^{-9}/2 [C/m]}{2\pi(2m) \cdot \frac{10^{-9}}{36\pi}} = \boxed{4.5 \hat{a}_r [V/m]}$$

$$(3) \vec{E}_B = \hat{a}_r \frac{\rho_l}{2\pi r \epsilon_0} \quad B(4m, \pi, 5m)$$

$$= \hat{a}_r \frac{\frac{10^{-9}}{2} [C/m]}{2\pi(4) \cdot \frac{10^{-9}}{36\pi}} = \boxed{\frac{2.25}{2.25} \hat{a}_r [V/m]}$$

$$(4) V_{AB} = V_B - V_A = - \int_B^A \frac{10^{-9}}{2 \cdot 2\pi\epsilon_0 r} dr = -9 [ln r]_4^2 = \boxed{6.24 V}$$

$$(b) \quad \vec{E} = \left(-\frac{16}{r^2}\right) \hat{a}_r \left[\frac{V}{m}\right] : \text{only as a function of } r$$

$$(1) \quad \vec{D} = \epsilon_0 \vec{E} = -\frac{16}{r^2 \epsilon_0} \hat{a}_r \left[\frac{C}{m}\right]$$

$$\vec{D}_B = -\frac{16}{(16)^2 \left(\frac{10^{-9}}{36\pi}\right)} \hat{a}_r = \frac{36\pi}{16 \times 10^{-9}} \hat{a}_r =$$

$$\begin{matrix} A(2m, \pi, \frac{\pi}{2}) \\ B(2m, \pi, \frac{\pi}{2}) \rightarrow (4m, 0, \pi) \end{matrix} \quad \boxed{2.25\pi \times 10^9 \hat{a}_r \left[\frac{C}{m}\right]}$$

$$(2) \quad V_{AB} = V_B - V_A = -\int_B^A \left(-\frac{16}{r^2}\right) dr = \boxed{-4V}$$

9.

$$(a) \quad \vec{E} = 0.15 MV/m, \quad N_e = 4.25. \quad \rightarrow N_e = \epsilon_r - 1 \quad \epsilon_r = 5.25$$

$$(1) \quad \vec{D} = \epsilon_0 \epsilon_r \vec{E} = \left(\frac{10^{-9}}{36\pi}\right) \cdot (5.25) \cdot 0.15 \times 10^6 = \boxed{6.97 \times 10^{-6} \left[\frac{C}{m^2}\right]}$$

$$(2) \quad \vec{P} = \epsilon_0 N_e \vec{E} = \left(\frac{10^{-9}}{36\pi}\right) \cdot (4.25) \cdot 0.15 \times 10^6 = \boxed{5.64 \times 10^{-6} \left[\frac{C}{m^2}\right]}$$

$$(3) \quad \epsilon_r = N_e + 1 = 4.25 + 1 = \boxed{5.25}$$

(b) From the definition,

$$\vec{P} = \epsilon_0 \chi_e \vec{E} \quad (\chi_e = 3.5)$$

$$\begin{aligned} \vec{E} &= \frac{1}{\epsilon_0 \chi_e} \vec{P} = \frac{1}{\left(\frac{10^{-9}}{36\pi}\right) \cdot (3.5)} (2.3 \times 10^{-7}) \hat{a} \left[ \frac{C}{m^2} \right] \\ &= \boxed{7.42 \times 10^3 \hat{a} \left[ \frac{V}{m} \right]} \end{aligned}$$

4.

(a) With fringing neglected Gauss's law as we discussed in chapters 3-4-7 & 3-4-8,

(1) At  $r=a$ , if  $\rho_s$  is surface charge at the surface of inner conductor

$$\vec{D} = \rho_s \frac{a}{r} \hat{a}_r \longrightarrow \vec{E} = \frac{\rho_s a}{\epsilon r} \hat{a}_r$$

Therefore,

$$V_{ba} = V_a - V_b = \int_a^b \left( \frac{\rho_s a}{\epsilon r} \right) \hat{a}_r \cdot \hat{a}_r dr = \frac{\rho_s a}{\epsilon} \ln\left(\frac{a}{b}\right)$$

(2)  $C = ?$  finding from  $Q = CV$

$Q =$  Total charge on the inner conductor

$$Q = \rho_s A = \rho_s 2\pi a h$$

$$V_{ba} = \frac{\rho_s}{\epsilon} \ln \frac{a}{b}, \quad C = \frac{Q}{V} = \frac{\rho_s 2\pi a h}{\frac{\rho_s}{\epsilon} \ln \left(\frac{a}{b}\right)} = \boxed{\frac{2\pi \epsilon h}{\ln(a/b)}}$$

(3)  $\vec{E}$ ,  $\vec{D}$ , & stored energy  $W_e$

$$W_e = \frac{1}{2} \int \vec{D} \cdot \vec{E} \, dv = \frac{1}{2} QV = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$$

$$W_e = \frac{1}{2} QV = \frac{1}{2} (\rho_s 2\pi a h) \cdot \left(\frac{\rho_s a}{\epsilon}\right) \ln\left(\frac{a}{b}\right)$$

$$= \boxed{\frac{1}{2} \frac{\rho_s^2 a^2 2\pi h}{\epsilon} \ln\left(\frac{a}{b}\right)}$$

(b) From the figure, it is understood that two capacitors are connected in parallel, hence

$$(i) C_{total} = C_1 + C_2$$

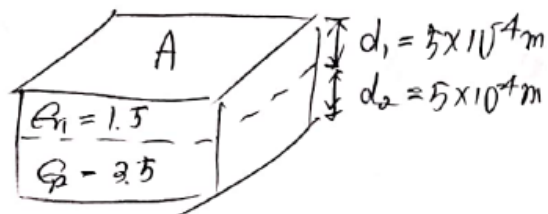
$$C_1 = \frac{\epsilon_0 \epsilon_{r1} A_1}{d}, \quad C_2 = \frac{\epsilon_0 \epsilon_{r2} A_2}{d}, \quad A_1 = A_2 = 1 \text{ m}^2$$

$$C_1 = \frac{\left(\frac{10^{-9}}{36\pi}\right) (1.5) \cdot 1}{10^{-3}} = 13.3 \text{ nF}$$

$$C_2 = \frac{\left(\frac{10^{-9}}{36\pi}\right) (3.5) \cdot 1}{10^{-3}} = 31.0 \text{ nF}$$

$$\text{Therefore, } \boxed{C_{total} = C_1 + C_2 = 44.3 \text{ nF}}$$

(ii)



Two capacitors are connected in series, thus,

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$$

$$C_1 = \frac{\epsilon_0 \epsilon_1 A}{d_1} = \frac{\left(\frac{10^{-9}}{36\pi}\right) (1.5) (2)}{5 \times 10^{-4}} = 53 \text{ nF}$$

$$C_2 = \frac{\epsilon_0 \epsilon_2 A}{d_2} = \frac{\left(\frac{10^{-9}}{36\pi}\right) (3.5) (2)}{5 \times 10^{-4}} = 123.7 \text{ nF}$$

Therefore,  $C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = 37.10 \text{ nF}$

5.

$$(a) \vec{E} = -\nabla V = -\left(\frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z\right) = -2\hat{a}_x - 4\hat{a}_y \left[\frac{V}{m}\right]$$

$$V = 2x + 4y$$

$$|\vec{E}| = \sqrt{(-2)^2 + (-4)^2} = \sqrt{20} \left[\frac{V}{m}\right]$$

$$W_e = \frac{1}{2} \int_V \epsilon E^2 dV$$

$$\text{Energy density, } w_e = \frac{1}{2} \epsilon E^2 = \frac{1}{2} (\sqrt{20})^2 \left(\frac{10^{-9}}{36\pi}\right) = \frac{10^{-8}}{36\pi} \text{ [J/m}^3\text{]}$$

Therefore, total volume is  $1 \text{ m}^3$

$$W_e = \left(\frac{1}{2} \epsilon E^2\right) \cdot V = \frac{10^{-8}}{36\pi} \times 1 = \frac{10^{-8}}{36\pi} \text{ [J]}$$

$$(b) \quad \vec{E} = \left( \frac{10^5}{r} \right) \hat{a}_r \text{ [V/m]} \text{ in cylindrical coordinates,}$$

$$W_e = \frac{1}{2} \int_V \epsilon_0 E^2 dV = \frac{\epsilon_0}{2} \int_V E^2 dV$$

$$= \frac{\epsilon_0}{2} \int_0^L \int_0^{2\pi} \int_{r_1}^{r_2} \left( \frac{10^5}{r} \right)^2 r dr d\phi dz$$

$$= \frac{\epsilon_0}{2} \int_h^{h+0.5} \int_0^{2\pi} \int_{0.01}^{0.05} \left( \frac{10^5}{r} \right)^2 r dr d\phi dz$$

$$= \boxed{0.224 \text{ [J]}}$$